

OMNIVERSE SET UP

AWS G5 · NVIDIA A10G · Omniverse Kit

AGENDA

1. Requesting GPU quota
2. Launch virtual machine (Azure vs AWS G5)
3. NVIDIA GPU driver
4. CUDA Toolkit
5. Omniverse Kit

REQUESTING QUOTA INCREASE

GPU quota is a **mandatory** step. Without it, **Launch instance** fails with a vCPU limit error.

AWS (target for this engagement)

Item	Value
Family	**G5** — NVIDIA **A10G** Tensor Core (24 GB VRAM)
Quota name	**Running On-Demand G and VT instances**
Unit	**vCPUs per Region** (not “number of VMs”)
Default	Often **0** on a new account
Where	AWS Console → **Service Quotas** → **Amazon EC2**
Filter	`On-Demand` + `G and VT`
Do not use	Spot quota (`All G and VT Spot Instance Requests`) — will not launch On-Demand G5

Recommended request (demo / workstation)

Use case	Instance	vCPUs	Why
Kit smoke test	`g5.2xlarge`	**8** (+ buffer)	Minimum comfortable Windows + Kit
RTX workstation (preferred)	`g5.4xlarge`	**16**	Closest AWS match to Azure `Standard_NV18ads_A10_v5`
Heavy USD / multi-viewport	`g5.8xlarge`	**32**	More host RAM and EBS / network

Always request **current usage + planned G5 vCPUs + 10–30 % buffer**, in the **same Region** where you will launch.

Quota is **regional**. Approving G5 in `us-east-1` does nothing for `us-west-2`. Pick a Region with G5 capacity before filing the ticket.

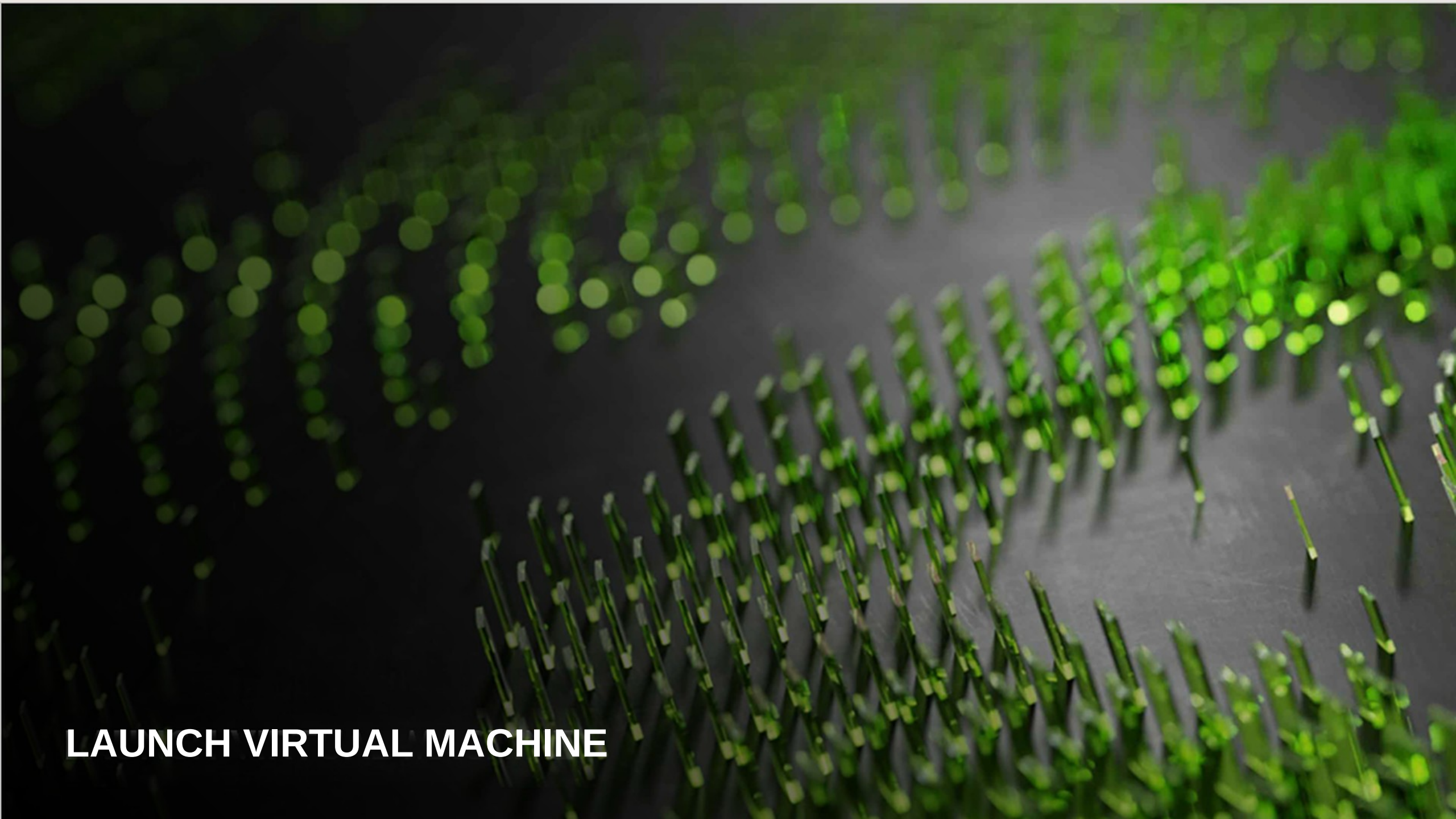
REQUESTING QUOTA INCREASE — AZURE PARALLEL

Azure (template parallel)

Item	Value
VM size	`Standard_NV18ads_A10_v5`
Need	**18 vCPUs** in the NV A10 v5 family
Where	Azure Portal → **Quotas** → Compute
Filter	`nvadsa10v5` · Region e.g. **East US**

Same rule, both clouds

- Quota is a **manual portal request** — not Terraform, CloudFormation, or Bicep.
- Unit is **vCPUs per Region**, never “number of VMs”.
- Request in the **same Region** you will launch in; approval is not global.
- File the ticket **before** the workshop — approval can take hours to days.
- Do not confuse **On-Demand** quota with **Spot** quota.


LAUNCH VIRTUAL MACHINE

LAUNCH AZURE VIRTUAL MACHINE

Connect to the virtual machine over **RDP**.

Typical training SKU from the template

- Size: `Standard_NV18ads_A10_v5`
- GPU: NVIDIA **A10**
- OS: Windows 11 Pro / Enterprise image
- Access: RDP **3389** only from corporate CIDR

Why it maps 1:1 to AWS G5

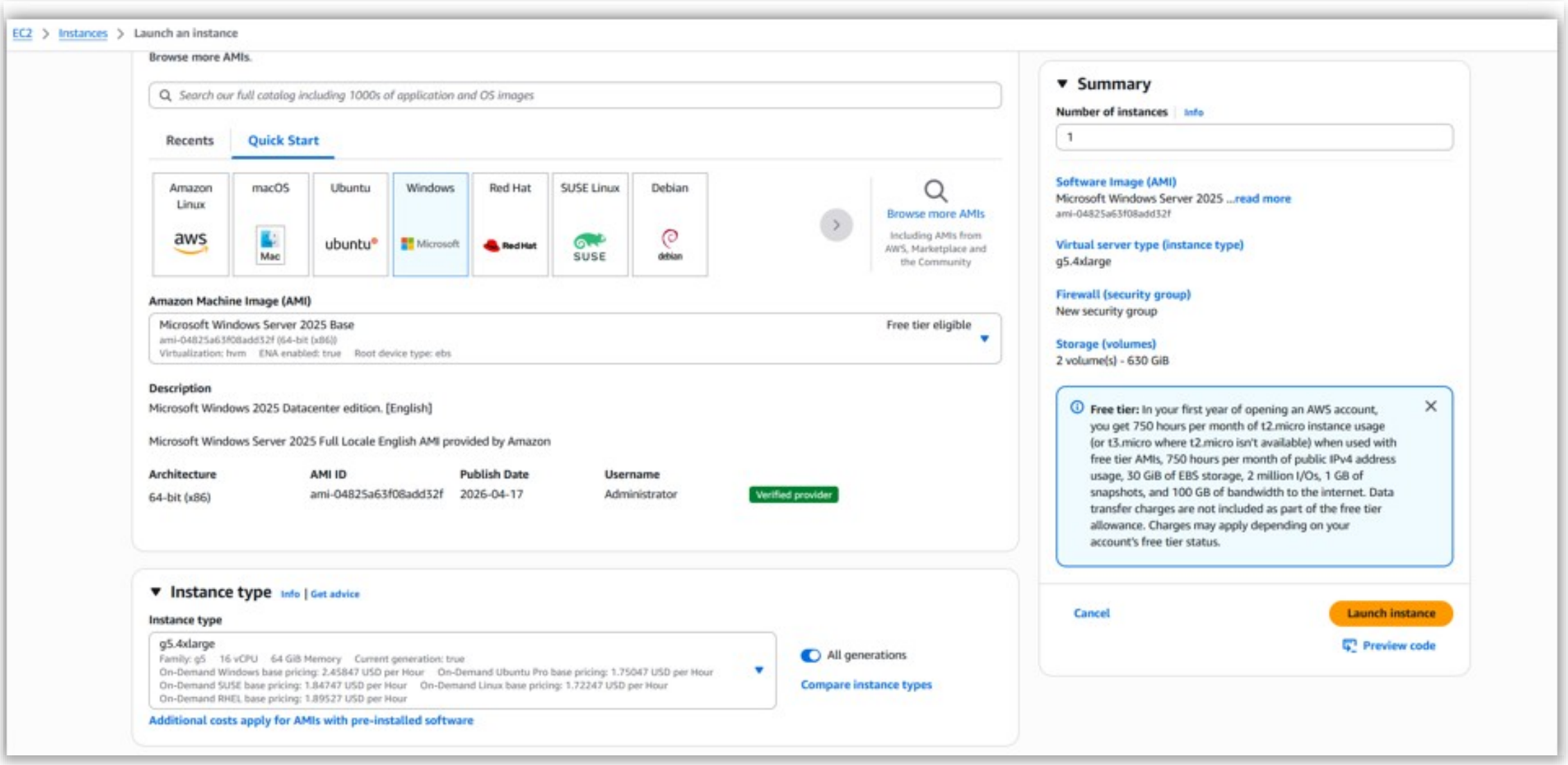
Two clouds, one GPU class: **NVIDIA Ampere A10 / A10G** for RTX. Everything after the VM launch (driver → CUDA → Kit) is identical on Azure and AWS.

LAUNCH AZURE VIRTUAL MACHINE — AWS MAPPING

Azure	AWS G5 equivalent
`Standard_NV18ads_A10_v5` (18 vCPU, A10)	**`g5.4xlarge`** (16 vCPU, 64 GiB, **1× A10G 24 GB**)
Resource group + VNet + NSG	VPC + subnet + **Security Group** (+ optional NACL)
Public IP + NIC	Elastic IP (only if required) + ENI
NVIDIA GPU Driver **Extension**	AWS **GRID / RTX vWS** driver (or NVIDIA install)
Quota: NV family vCPUs	Quota: **G and VT** On-Demand vCPUs

A10 vs A10G: same Ampere generation, 24 GB VRAM. A10G is the AWS data-center SKU. RTX in Omniverse is supported on both.

LAUNCH AWS INSTANCE



LAUNCH AWS INSTANCE — G5 FAMILY (NVIDIA A10G)

Instance	GPUs	vCPU	RAM	Local NVMe	When to use
g5.xlarge	1	4	16 GiB	250 GB	Too tight for Windows + Kit — avoid
g5.2xlarge	1	8	32 GiB	450 GB	Demo / Kit launch
g5.4xlarge	1	16	64 GiB	600 GB	**Recommended workstation**
g5.8xlarge	1	32	128 GiB	900 GB	Large USD assemblies
g5.16xlarge	1	64	256 GiB	1900 GB	Single GPU, max host RAM
g5.12xlarge	4	48	192 GiB	3800 GB	Multi-GPU (not needed for first Kit)
g5.24xlarge	4	96	384 GiB	3800 GB	Farm / batch
g5.48xlarge	8	192	768 GiB	7600 GB	Max A10G density

All sizes: NVIDIA **A10G** Tensor Core, 24 GB GDDR6 per GPU, RTX / ray tracing capable.

LAUNCH AWS INSTANCE — EC2 CONSOLE CHECKLIST

1. **AMI**
   - Default marketplace: **Windows Server 2022 / 2025**.
   - **Windows 11** is not a stock G5 AMI — import a custom VHD / AMI (BYOL) if the customer mandates Win11.
   - Linux / headless Kit or Isaac: **NVIDIA Omniverse GPU-Optimized AMI** (`g5.2xlarge`+).
1. **Instance type:** `g5.4xlarge` (preferred) or `g5.2xlarge`.
2. **Subnet:** private preferred; jump host or VPN for RDP.
4. **Storage:** gp3 root **≥ 256 GB** (Kit + CUDA + USD). Instance store is ephemeral — do not keep projects only there.
5. **IAM:** SSM Session Manager role for console access without opening 3389 to the internet.
6. **Key pair:** required for Windows password decrypt (unless SSM + AD).

Network controls

Security Groups apply at the instance (ENI). **NACLs** apply at the subnet. Use the SG as the primary control and the NACL as a coarse extra layer.

LAUNCH AWS INSTANCE — SECURITY GROUP

Direction	Port	Source / Destination	Purpose
Inbound	TCP **3389**	Corporate CIDR only	RDP
Inbound	TCP **22**	Bastion / VPN only	Linux SSH (if used)
Inbound	TCP **443**	VPC / VPN	Nucleus / HTTPS if co-located
Outbound	HTTPS **443**	`0.0.0.0/0`	CUDA, NGC, GitHub, Windows Update

Never open RDP to `0.0.0.0/0`. Prefer **VPN / Direct Connect + private IP**, or SSM Session Manager port-forwarding, with an MFA-protected jump host.

LAUNCH AWS INSTANCE — DRIVERS ON G5

Two valid paths:

1 · AWS GRID driver (recommended for Windows RTX / Omniverse desktop)

- Follow **Install NVIDIA drivers on Windows** for **G5** (AWS docs).
- AWS publishes PowerShell helpers that pull the GRID package from S3.
- Reboot after install, then verify with `nvidia-smi`.

```
# PowerShell (Administrator) - G5 GRID driver
$Bucket    = "ec2-windows-nvidia-drivers"
$KeyPrefix = "latest"
$Objects = Get-S3Object -BucketName $Bucket `
             -KeyPrefix $KeyPrefix -Region us-east-1
foreach ($o in $Objects) {
  Copy-S3Object -BucketName $Bucket -Key $o.Key `
    -LocalFile "$home\Desktop\NVIDIA\$($o.Key)" `
    -Region us-east-1
}
# run the downloaded installer, then Restart-Computer
```

2 · Manual NVIDIA package inside the OS

- Same idea as the Azure “optional local install” slide.
- Download the **RTX Virtual Workstation (vWS)** / GRID branch for A10G — not Game Ready.
- Requires an NVIDIA account for the download.
- Reboot, then `nvidia-smi` must list **NVIDIA A10G**.

Keep **one** known-good combination per golden image: **G5 GRID driver + CUDA 13.2 + Kit build**. Snapshot the AMI once all three work.

LAUNCH VIRTUAL MACHINE WITH NVIDIA GPU DRIVER

Cloud	Mechanism
Azure	VM extension **NVIDIA GPU Driver** (`Microsoft.HpcCompute` / `NvidiaGpuDriverWindows`)
AWS G5	**GRID / RTX Virtual Workstation** driver for graphics + RTX

- Azure SKU in the original deck: `Standard_NV18ads_A10_v5` — gen-5 **A10**, RTX-capable.
- AWS SKU for the same class of work: **`g5.4xlarge`** — **A10G**, 24 GB.

NVIDIA GPU DRIVER

- Optional / parallel path: install the NVIDIA GPU driver **inside** the VM (not only via the Azure extension).
- On AWS: after the GRID install, **reboot**, then verify with `nvidia-smi`.
- Pick the **graphics** branch (GRID / RTX vWS). A Tesla compute-only driver will not give you an RTX viewport.

Expected nvidia-smi header on a g5.4xlarge

```
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 5xx.xx        Driver Version: 5xx.xx        CUDA Version: 13.2               |
|-----------------------------------------+-------------------------+---------------------+
| GPU  Name                 Driver-Model | Bus-Id          Disp.A | Volatile Uncorr. ECC |
|   0  NVIDIA A10G                 WDDM  | 00000000:00:1E.0    On |                    0 |
+-----------------------------------------+-------------------------+---------------------+
```

INSTALL NVIDIA DRIVER

Use the **NVIDIA / AWS GRID driver first**, then confirm with **`nvidia-smi`**.

Verify

1. RDP to the instance.
2. Open **Command Prompt** or **PowerShell as Administrator**.
3. Run:

```
cd "C:\Program Files\NVIDIA Corporation\NVSMI"
nvidia-smi
```

If `NVSMI` is not on that path (newer driver layouts), it is already on `PATH`:

```
nvidia-smi
```

Expected

- GPU name: **NVIDIA A10G** (AWS) or **A10** (Azure)
- Driver version listed
- `GPU-Util` may be `N/A` or `0%` with no workload — that is normal

If nvidia-smi fails

Wrong driver branch (Tesla compute-only vs GRID graphics), incomplete install, or **no reboot**.


INSTALL CUDA TOOLKIT

INSTALL CUDA TOOLKIT

CUDA Toolkit is required for RTX rendering, physics, and GPU compute workloads.

Prerequisite: NVIDIA account — sign-in is required to download the package.

Reference

developer.nvidia.com/cuda-downloads (Windows · x86_64 · exe local)

Download

1. Open the CUDA downloads page: **Windows · x86_64 · Windows 11 or Windows Server · exe (local)**.
2. Download **CUDA Toolkit 13.2** (same pin as the demo template).
3. Run the installer **as Administrator**. Reboot if requested.

Ops note

The download is a **manual browser action** behind an NVIDIA login. Do not expect this step to be automated in IaC — bake it into the golden AMI instead.

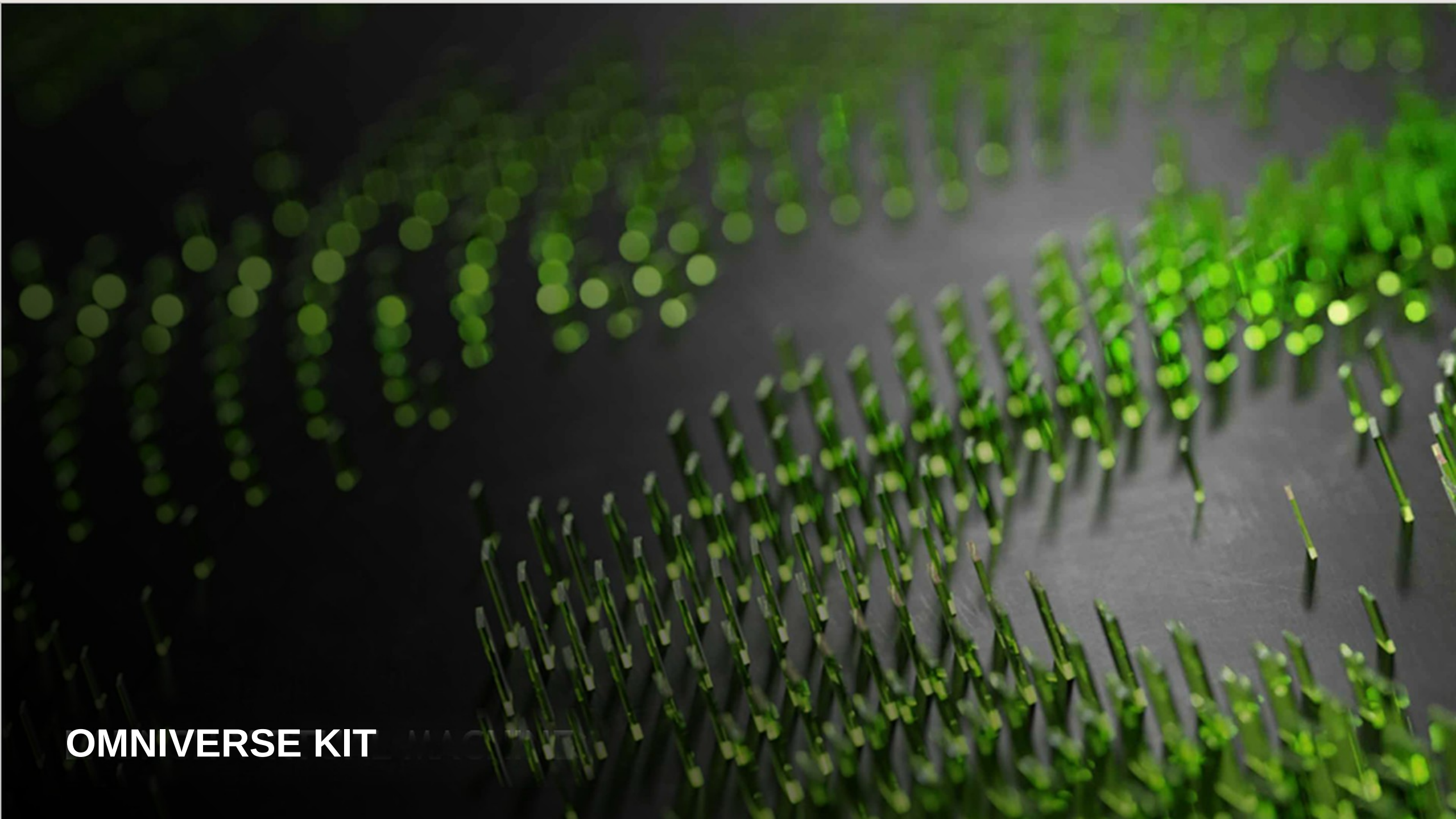
INSTALL CUDA TOOLKIT — VERIFY

After install, confirm compiler and driver still agree:

```
nvidia-smi
where nvcc
nvcc --version
```

- `nvidia-smi` → driver version and the **CUDA version the driver supports**
- `where nvcc` → `C:\Program Files\NVIDIA GPU Computing Toolkit\CUDA\v13.2\bin\nvcc.exe`
- `nvcc --version` → **release 13.2**

Keep **one** known-good combo: **G5 GRID driver + CUDA 13.2 + Kit build**. Do not mix random Game Ready drivers on a cloud GPU.


OMNIVERSE KIT

LAUNCH OMNIVERSE KIT

1 · Clone NVIDIA Omniverse kit-app-template

```
git clone https://github.com/NVIDIA-Omniverse/kit-app-template.git
cd kit-app-template
```

2 · Create a new application from template

```
.\repo.bat template new
```

3 · Build

```
.\repo.bat build
```

4 · Launch

```
.\repo.bat launch
```

Requires **Git for Windows** and a working GPU driver (`nvidia-smi` OK) before first launch. First build downloads the Kit SDK from NGC — allow outbound 443.

LAUNCH OMNIVERSE KIT — FIRST-RUN WIZARD

Complete the **initial setup** of the Kit application:

1. Accept **license and consent**.
2. **Create Application**.
3. Set:
   - application name
   - `.kit` file name
   - display name
   - application version
4. Leave remaining options at **defaults** for the first setup.

Recommendation

Defaults are recommended for a first successful launch. Customise the `.kit` file and extensions only after the UI has loaded once.

LAUNCH OMNIVERSE KIT — UI LOADED

UI load confirms:

- the application was **created successfully**
- the **Kit runtime** is working
- the environment is ready for **USD scenes, extensions, and RTX**
- next step: connect to **Enterprise Nucleus** for collaboration

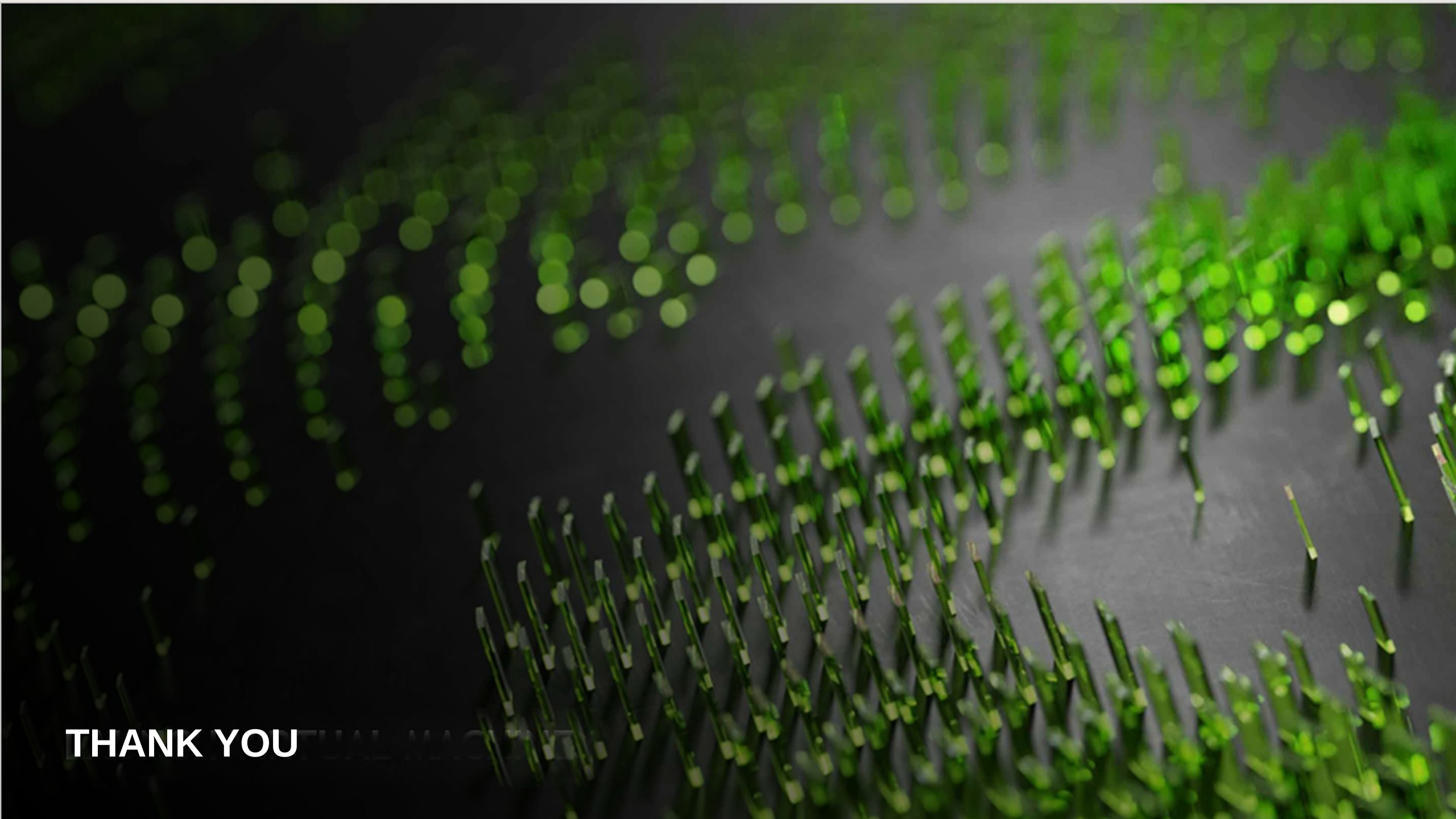
Nucleus placement

Nucleus is a **separate service** — typically a CPU instance with a persistent EBS volume and backups. Do not co-host it on the GPU workstation except for a tiny lab.

REFERENCES

REFERENCES

Topic	Link
CUDA downloads (Win x64, local exe)	https://developer.nvidia.com/cuda-downloads?target_os=Windows&target_arch=x86_64&target_version=11&target_type=exe_local
Omniverse Kit (NGC)	https://catalog.ngc.nvidia.com/orgs/nvidia/teams/omniverse/collections/kit
Enterprise Nucleus Server	https://catalog.ngc.nvidia.com/orgs/nvidia/teams/omniverse/collections/enterprise-nucleus/artifacts
kit-app-template	https://github.com/NVIDIA-Omniverse/kit-app-template
EC2 G5 (A10G)	https://aws.amazon.com/ec2/instance-types/g5/
On-Demand G / VT quota	https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-on-demand-instances.html
NVIDIA drivers on Windows EC2	https://docs.aws.amazon.com/AWSEC2/latest/WindowsGuide/install-nvidia-driver.html
Azure N-series Windows drivers	https://learn.microsoft.com/en-us/azure/virtual-machines/windows/n-series-driver-setup


THANK YOU

APPENDIX — RUNNING THIS IN PRODUCTION

- **Quota first**, then AMI, then instance. IaC cannot skip Service Quotas.
- **Pin** Region + AZ with G5 capacity; consider a **Capacity Reservation** for a workshop date.
- **Stop** G5 when idle — A10G On-Demand is expensive. Instance store is wiped on stop — keep USD on **EBS**.
- **Nucleus** on a non-GPU instance with backups; Kit clients on G5.
- **Do not** put RDP on a public IP without CIDR lock + MFA jump host.
- Snapshot / AMI the golden image **after** GRID + CUDA 13.2 + Kit build succeed.

APPENDIX — FIRST-RUN VALIDATION CHECKLIST

- ☐ `Running On-Demand G and VT instances` quota ≥ instance vCPUs in the target Region
- ☐ Instance is `g5.2xlarge` or `g5.4xlarge` and `running`
- ☐ SG: RDP only from corp / VPN
- ☐ GRID driver installed; reboot done
- ☐ `nvidia-smi` shows **A10G**
- ☐ CUDA Toolkit **13.2**; `nvcc --version`
- ☐ `kit-app-template` cloned; `.\repo.bat build` / `launch` succeeds
- ☐ Kit UI loads; RTX viewport available